CStats - Lab #6

March 31, 2026

Exercise 1.
An engineer is interested in the effects of cutting speed (A), tool geometry (B), and cutting angle (C) on the life (in hours) of a machine tool. Two levels of each factor are chosen, and three replicates of a 2^3 factorial design are run. The results are as in Table 1.
(a) Estimate the factor effects. Which effects appear to be large?
(b) Use the analysis of variance to confirm your conclusions for part **(a)**.
(c) Write down a regression model for predicting tool life (in hours) based on the results of this experiment.
(d) Analyze the residuals. Are there any obvious problems?
(e) On the basis of an analysis of main effect and interaction, what factor levels of A, B, and C would you recommend using?

A	B	C	Treatment Combination	Replicate		
				I	II	III
−	−	−	(1)	22	31	25
+	−	−	*a*	32	43	29
−	+	−	*b*	35	34	50
+	+	−	*ab*	55	47	46
−	−	+	*c*	44	45	38
+	−	+	*ac*	40	37	36
−	+	+	*bc*	60	50	54
+	+	+	*abc*	39	41	47

Table 1: Treatment Combinations and Results

Solution 1.

(a) At first, let calculate the total life of a machine tool by averaging all replicates. The column of the mean by treatment combinations is as the following:

$$(1) = 26$$
$$a = 34.67$$
$$b = 39.67$$
$$ab = 49.33$$
$$c = 42.33$$
$$ac = 37.67$$
$$bc = 54.67$$
$$abc = 42.33$$

To calculate the factor effects, we use the equations for the model parameters such that $q_j = \frac{1}{2^k}\sum_i s_{ij}\bar{y}_i$. We have $n = 3$ as the number of replicates and $N = 24$ as the number of observations.

$$q_A = \frac{1}{2^k}(a + ab + ac + abc - (1) - b - c - bc) = 0.17$$
$$q_B = \frac{1}{2^k}(b + ab + bc + abc - (1) - a - c - ac) = 5.67$$
$$q_C = \frac{1}{2^k}(c + ac + bc + abc - (1) - a - b - ab) = 3.41$$
$$q_{AB} = \frac{1}{2^k}(ab + abc + (1) - a - b + c - ac - bc) = -0.83$$
$$q_{AC} = \frac{1}{2^k}((1) - a + b - ab + abc - c + ac - bc) = -4.42$$
$$q_{BC} = \frac{1}{2^k}((1) + a - b - ab - c - ac + bc + abc) = -1.42$$
$$q_{ABC} = \frac{1}{2^k}(abc - bc - ac + c - ab + b + a - (1)) = -1.08$$

We note that the overall mean q_0 is 40.83. **(b)** Then, using $SS_i = 2^k r q_j^2$, we can calculate sum of squares as the following:

$$SS_A = 0.66$$
$$SS_B = 770.67$$
$$SS_C = 280.17$$
$$SS_{AB} = 16.67$$
$$SS_{AC} = 468.17$$
$$SS_{BC} = 48.17$$
$$SS_{ABC} = 28.17$$

Sum of squares of the model is

$$SS_M = SS_A + SS_B + SS_C + SS_{AB} + SS_{AC} + SS_{BC} + SS_{ABC} = 1612.66$$

The total sum of squares is:

$$SST = SSY - SS0 = 2095.33$$

Therefore, sum of squares for the error is

$$SSE = SST - SS_M = 482.67$$

We are ready to calculate the mean squares. Let define degrees of freedoms as the following and calculate them.

$$\begin{aligned}
df_A &= 1 \\
df_B &= 1 \\
df_C &= 1 \\
df_{AB} &= 1 \\
df_{AC} &= 1 \\
df_{BC} &= 1 \\
df_{ABC} &= 1 \\
df_T &= N - 1 = 23 \\
df_E &= df_T - (df_A + df_B + df_C + df_{AB} + df_{AC} + df_{BC} + df_{ABC}) = 16 \\
df_M &= 7
\end{aligned}$$

where df_M is degrees of freedom of this design.

$$\begin{aligned}
MS_A &= SS_A/df_A = 0.66 \\
MS_B &= SS_B/df_B = 770.67 \\
MS_C &= SS_C/df_C = 280.17 \\
MS_{AB} &= SS_{AB}/df_{AB} = 16.67 \\
MS_{AC} &= SS_{AC}/df_{AC} = 468.17 \\
MS_{BC} &= SS_{BC}/df_{BC} = 48.17 \\
MS_{ABC} &= SS_{ABC}/df_{ABC} = 28.17 \\
MSE &= SSE/df_E = 30.17
\end{aligned}$$

Thus, F-statistics for each factor is:

$$\begin{aligned}
F_A &= MS_A/MSE = 0.02 \\
F_B &= MS_B/MSE = 25.54 \\
F_C &= MS_C/MSE = 9.29 \\
F_{AB} &= MS_{AB}/MSE = 0.55 \\
F_{AC} &= MS_{AC}/MSE = 15.51 \\
F_{BC} &= MS_{BC}/MSE = 1.59 \\
F_{ABC} &= MS_{ABC}/MSE = 0.93
\end{aligned}$$

Then, we can calculate the p-values:

$$\begin{aligned}
p_A &= 1 - F_{(F_A, df_A, df_E)} = 0.88 \\
p_B &= 1 - F_{(F_B, df_B, df_E)} = 0.0001 \\
p_C &= 1 - F_{(F_C, df_C, df_E)} = 0.007 \\
p_{AB} &= 1 - F_{(F_{AB}, df_{AB}, df_E)} = 0.46 \\
p_{AC} &= 1 - F_{(F_{AC}, df_{AC}, df_E)} = 0.001 \\
p_{BC} &= 1 - F_{(F_{BC}, df_{BC}, df_E)} = 0.22 \\
p_{ABC} &= 1 - F_{(F_{ABC}, df_{ABC}, df_E)} = 0.34
\end{aligned}$$

The factor effects which have $p < 0.05$ is significant for the model. As we see the p-values, the factors B and C and interaction of A and C is significant.
(c) The model is expressed by

$$\hat{y} = q_0 + q_B x_B + q_C x_C + q_{AC} x_A x_C$$

(d) We already have the results for residuals (error) from previous part. Let recall them:

$$\begin{aligned}
SSE &= 482.67 \\
MSE &= 30.17
\end{aligned}$$

You can use Python to plot your residuals and explain your results. **(e)** Based on the results from part (b), the factors B and C can be recommended.

Exercise 2.

The effect estimates from a 2^4 factorial experiment are listed here. Let say there is 2 replicates in this experiment. Can we decide which factors are significant for $\alpha = 0.05$ level? If yes, write a regression model using only significant factors.

$ABCD = -2.5251,$

$BCD = 4.4054, ACD = -0.4932, ABD = -5.0842, ABC = -5.7696,$

$CD = 4.6707, BD = -4.6620, BC = -0.7982, AD = -1.6564, AC = 1.1109, AB = -10.5229,$

$D = -6.0275, C = -8.2045, B = -6.5304, A = -0.7914$

Solution 2.
To definitively decide which factors are significant, we need the mean square error (MSE) from the residuals of the analysis and the degrees of freedom for error to calculate the standard errors of the effect estimates. Once these are known, we could perform hypothesis tests for each effect estimate and compare them to the p-value to assess statistical significance. Without the MSE or any other form of experimental variability information, it is not possible to determine definitively which factors are significant based solely on the effect estimates.

Exercise 3.
Is it possible to have a 2^{4-1}_{III} design? A 2^{4-1}_{IV} design? If yes, give an example. If no, explain why?

Solution 3.
A 2^{4-1}_{III} design is a Resolution III design with 4 factors and 8 runs. (I = ABD, I = ABC, ...) Main effects are aliased with 2-factor interactions.

To construct an example, the followings should be satisfied:

- The sum of each column is zero
- The sum of the products of any two columns is zero
- The sum of the squares of each column is 2^{k-p}

Here is an example of such a design:

A	B	C	D
-1	-1	-1	-1
-1	-1	$+1$	$+1$
-1	$+1$	-1	$+1$
-1	$+1$	$+1$	-1
$+1$	-1	-1	$+1$
$+1$	-1	$+1$	-1
$+1$	$+1$	-1	-1
$+1$	$+1$	$+1$	$+1$

This is a half-fraction of a 2^4 design, which means 2^3. The aliasing in Resolution III allows main effects to be aliased/confounded with two-factor interactions. (Confoundings: Only the combined influence of two or more effects can be computed.)

A 2^{4-1}_{IV} design is also possible because it requires 8 runs. I=ABCD. Main effects are only aliased with 3-factor interactions. 2-factor interactions are aliased with other 2-factor interactions

Exercise 4.

The researchers use 2^{5-2} design to investigate the effect of A condensation temperature, B amount of material 1, C solvent volume, D condensation time, and E amount of material 2 on yield. The results obtained are as follows:

$$e = 23.2, ad = 16.9, cd = 23.8, bde = 16.8$$
$$ab = 15.5, bc = 16.2, ace = 23.4, abcde = 18.1$$

(a) Verify that the design generators used were $I = ACE$ and $I = BDE$.
(b) Write down the complete defining relation and the aliases for this design.
(c) Estimate the main effects.

Solution 4.

(a) 2^{5-2} fractional factorial design is given. It means a design with 5 factors and 8 (2^3 design) runs. This implies that two factors are assigned based on generators. The given design suggests that factors A, B, C, D, E exist, and two generators, which are $I = ACE, I = BDE$, are used to define the fraction.

	A	B	C	D	E	ACE	BDE
e	$-$	$-$	$-$	$-$	$+$	$+$	$+$
ad	$+$	$-$	$-$	$+$	$-$	$+$	$+$
cd	$-$	$-$	$+$	$+$	$-$	$+$	$+$
bde	$-$	$+$	$-$	$+$	$+$	$+$	$+$
ab	$+$	$+$	$-$	$-$	$-$	$+$	$+$
bc	$-$	$+$	$+$	$-$	$-$	$+$	$+$
ace	$+$		$+$	$-$	$+$	$+$	$+$
$abcde$	$+$	$+$	$+$	$+$	$+$	$+$	$+$

From above, product of the three columns A, C, E and B, D, E gives all $+$. Therefore, $I = ACE, I = BDE$.

(b) Since $ACE \times BDE = ABCD$, $I = ACE = BDE = ABCD$. Then, we can write the followings:

- Using $I = ACE$:

$$\begin{aligned} A &= CE \\ B &= ABCE \\ C &= AE \\ D &= ACDE \\ E &= AC \\ AB &= BCE \\ AD &= CDE \end{aligned}$$

- Using $I = BDE$:

$$
\begin{aligned}
A &= ABDE \\
B &= DE \\
C &= BCDE \\
D &= BE \\
E &= BD \\
AB &= ADE \\
AD &= ABE
\end{aligned}
$$

- Using $I = ABCD$:

$$
\begin{aligned}
A &= BCD \\
B &= ACD \\
C &= ABD \\
D &= ABC \\
E &= ABCDE \\
AB &= CD \\
AD &= BC
\end{aligned}
$$

Therefore, when we bring all together, we get:

$$
\begin{aligned}
A &= CE = ABDE = BCD \\
B &= ABCE = DE = ACD \\
C &= AE = BCDE = ABD \\
D &= ACDE = BE = ABC \\
E &= AC = BD = ABCDE \\
AB &= BCE = ADE = CD \\
AD &= CDE = ABE = BC
\end{aligned}
$$

(c) Using the sign table in part **(a)**, we can write the followings:

$$
\begin{aligned}
q_A &= \frac{1}{2^k}[-e + ad - cd - bde + ab - bc + ace + abcde] \\
q_B &= \frac{1}{2^k}[-e - ad - cd + bde + ab + bc - ace + abcde] \\
q_C &= \frac{1}{2^k}[-e - ad + cd - bde - ab + bc + ace + abcde] \\
q_D &= \frac{1}{2^k}[-e + ad + cd + bde - ab - bc - ace + abcde] \\
q_E &= \frac{1}{2^k}[e - ad - cd + bde - ab - bc + ace + abcde]
\end{aligned}
$$